Sampling & Estimation

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Unbiased Estimators of Population Mean & Variance**

A random sample of 10 values of a variable x is taken. The results are summarised as follows: $\sum x = 50, \quad \sum x^2 = 322$ Calculate an unbiased estimate of the population variance.

To find an unbiased estimate of the population variance s^2, we use the formula: $s^2 = \frac{1}{n-1}\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)$ Given $n = 10$, $\sum x = 50$, and $\sum x^2 = 322$: $\frac{(\sum x)^2}{n} = \frac{50^2}{10} = \frac{2500}{10} = 250$ Substitute these values into the formula: $s^2 = \frac{1}{10-1}(322 - 250) = \frac{1}{9} \times 72 = 8$ - The distractor 7.2 results from dividing by $n = 10$ instead of $n - 1 = 9$, which gives the biased sample variance rather than the unbiase.

2 **[Foundational] • Target: The Central Limit Theorem (CLT)**

A random sample of $n = 50$ observations is taken from a population. The sample values are summarised as follows: $\sum x = 450, \quad \sum x^2 = 4295.$ Find an unbiased estimate of the population variance.

The unbiased estimate of the population variance, denoted s^2, is calculated using the formula: $s^2 = \frac{1}{n-1}\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)$ Substitute the given values $n = 50$, $\sum x = 450$, and $\sum x^2 = 4295$: $s^2 = \frac{1}{50-1}\left(4295 - \frac{450^2}{50}\right)$ $s^2 = \frac{1}{49}\left(4295 - \frac{202500}{50}\right)$ $s^2 = \frac{1}{49}(4295 - 4050) = \frac{245}{49} = 5.0$ Why other choices fail: - 4.9 is the biased sample variance obtained by dividing by.

3 **[Foundational] • Target: Confidence Intervals for the Population Mean**

The random variable X has a Poisson distribution with mean $\lambda > 0$. Given that $3\,\mathrm{P}(X = 2) = \mathrm{P}(X = 4)$, find the value of λ.

For a Poisson distribution $X \sim \mathrm{Po}(\lambda)$, the probability mass function is given by: $\mathrm{P}(X = r) = \frac{e^{-\lambda}\lambda^r}{r!}$ Substitute $r = 2$ and $r = 4$ into the given equation $3\,\mathrm{P}(X = 2) = \mathrm{P}(X = 4)$: $3\left(\frac{e^{-\lambda}\lambda^2}{2!}\right) = \frac{e^{-\lambda}\lambda^4}{4!}$ Since $e^{-\lambda} > 0$ and $\lambda > 0$, divide both sides by $e^{-\lambda}\lambda^2$: $\frac{3}{2} = \frac{\lambda^2}{24}$ Multiply both sides by 24: $\lambda^2 = 36$ Since.

4 **[Foundational] • Target: Confidence Intervals for Large Unknown-Variance Samples**

A curve satisfies the differential equation $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{k-y}{2x+1}$ for $x > -\frac{1}{2}$, where k is a real constant. The curve passes through the points $(0, 8)$ and $(4, 2)$. Find the value of k.

To find the constant k, we solve the differential equation by separation of variables: $\int \frac{1}{k-y}\,\mathrm{d}y = \int \frac{1}{2x+1}\,\mathrm{d}x$ Integrating both sides (paying close attention to the chain rule): $-\ln|k-y| = \frac{1}{2}\ln(2x+1) + C$ Multiplying by -1 and exponentiating: $\ln|k-y| = -\frac{1}{2}\ln(2x+1) - C = \ln\left((2x+1)^{-1/2}\right) - C$ $|k-y| = \frac{A}{\sqrt{2x+1}}$ where $A = \mathrm{e}^{-C} > 0$ Removing the absolute value gives th.

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5 [Foundational] • Target: Unbiased Estimators of Population Mean & Variance

Solve the equation $\ln(3x+1) - 2\ln 2 = \ln(x-1)$.

Using the power law of logarithms on the second term: $2\ln 2 = \ln(2^2) = \ln 4$. Substitute this back into the equation: $\ln(3x+1) - \ln 4 = \ln(x-1)$. Apply the quotient rule for logarithms, $\ln A - \ln B = \ln\left(\frac{A}{B}\right)$: $\ln\left(\frac{3x+1}{4}\right) = \ln(x-1)$. Since the natural logarithm function is one-to-one, we equate the arguments: $\frac{3x+1}{4} = x-1$. Multiply both sides by 4: $3x+1 = 4(x-1)$, $3x+1 = 4x-4$. Rearranging gives.

6 [Foundational] • Target: The Central Limit Theorem (CLT)

A curve has equation $y = x^2e^{3x}$. Find the gradient of the curve at the point where $x = 1$.

To find the gradient of the curve, differentiate $y = x^2e^{3x}$ with respect to x using the product rule: $\frac{\mathrm{d}}{\mathrm{d}x}[uv] = u\frac{\mathrm{d}v}{\mathrm{d}x} + v\frac{\mathrm{d}u}{\mathrm{d}x}$ Let $u = x^2$ and $v = e^{3x}$. Then $\frac{\mathrm{d}u}{\mathrm{d}x} = 2x$ and, by the chain rule, $\frac{\mathrm{d}v}{\mathrm{d}x} = 3e^{3x}$. Applying the product rule: $\frac{\mathrm{d}y}{\mathrm{d}x} = x^2(3e^{3x}) + 2x(e^{3x}) = (3x^2+2x)e^{3x}$ Substitute $x = 1$.

7 [Foundational] • Target: Confidence Intervals for the Population Mean

The continuous function f is defined for all $x \in \mathbb{R}$ by $f(x) = 2^x + 3x - 11$. Between which pair of consecutive integers does a root of the equation $f(x) = 0$ lie?

To determine the interval containing a root of a continuous function $f(x)$, we use the change-of-sign rule (Intermediate Value Theorem). If $f(a)$ and $f(b)$ have opposite signs, then at least one real root of $f(x) = 0$ lies in the interval (a, b). Evaluate $f(x)$ at consecutive integers: - For $x = 0$:
$f(0) = 2^0 + 3(0) - 11 = 1 + 0 - 11 = -10 < 0$ - For $x = 1$:
$f(1) = 2^1 + 3(1) - 11 = 2 + 3 - 11 = -6 < 0$ - For $x = 2$:
$f(2) = 2^2 + 3(2) - 11 = 4 + 6 - 11 = -1 < 0$ - For $x = 3$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Confidence Intervals for Large Unknown-Variance Samples

The first three terms of a geometric progression are given by $u_1 = 3k - 2, \quad u_2 = k + 2, \quad u_3 = k - 2$, where k is a real constant. Given that the infinite geometric series $\sum_{n=1}^{\infty} u_n$ converges, find the sum to infinity, S_∞.

Step 1: Set up the common ratio property of a geometric sequence. Since the terms form a geometric progression, the ratio between consecutive terms must be equal: $\frac{u_2}{u_1} = \frac{u_3}{u_2} \Rightarrow u_2^2 = u_1u_3$ Step 2: Substitute the expressions in terms of k: $(k+2)^2 = (3k-2)(k-2)$ $k^2 + 4k + 4 = 3k^2 - 8k + 4$ Step 3: Solve the quadratic equation for k: $2k^2 - 12k = 0 \Rightarrow 2k(k-6) = 0$ Thus, $k = 0$ or $k = 6$. Step 4: Apply the convergence criterion $|r| < 1$: - If $k = 0$.

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9 [Intermediate] • Target: Unbiased Estimators of Population Mean & Variance

Relative to a fixed origin O, the position vectors of the points A and B are given by $\vec{OA} = \mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$ and $\vec{OB} = 4\mathbf{i} + 2\mathbf{j} - 8\mathbf{k}$. Find the unit vector in the direction of $\vec{AB}$.

First, find the displacement vector $\vec{AB}$: $\vec{AB} = \vec{OB} - \vec{OA} = (4-1)\mathbf{i} + (2-(-2))\mathbf{j} + (-8-4)\mathbf{k} = 3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k}$. Next, calculate the magnitude of $\vec{AB}$: $|\vec{AB}| = \sqrt{3^2 + 4^2 + (-12)^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$. The unit vector in the direction of $\vec{AB}$ is given by $\frac{\vec{AB}}{|\vec{AB}|}$: $\frac{1}{13}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$. Common misconceptions: - Reversing the subtraction (cal.

10 [Intermediate] • Target: The Central Limit Theorem (CLT)

Find the exact value of the definite integral $\int_0^4 \frac{1}{2x+1}\,\mathrm{d}x$.

To evaluate the definite integral $\int_0^4 \frac{1}{2x+1}\,\mathrm{d}x$: 1. Use the standard integration rule $\int \frac{1}{ax+b}\,\mathrm{d}x = \frac{1}{a}\ln|ax+b| + C$. Here, $a = 2$ and $b = 1$, giving the antiderivative: $\int \frac{1}{2x+1}\,\mathrm{d}x = \frac{1}{2}\ln|2x+1|$ 2. Evaluate between the limits $x = 0$ and $x = 4$: $\left[\frac{1}{2}\ln|2x+1|\right]_0^4 = \frac{1}{2}\ln(2(4)+1) - \frac{1}{2}\ln(2(0)+1) = \frac{1}{2}\ln 9 - \frac{1}{2}\ln 1$ 3. Using the propert.

11 [Intermediate] • Target: Confidence Intervals for the Population Mean

The continuous random variable X has probability density function given by $f(x) = \{k(4-x^2), 0 \le x \le 2, ; 0, \text{otherwise},$ where k is a constant. Find the value of k.

For $f(x)$ to be a valid probability density function, the total area under the curve must equal 1: $\int_{-\infty}^{\infty} f(x)\,dx = 1$ Substitute the non-zero definition of $f(x)$: $\int_0^2 k(4-x^2)\,dx = 1$ Factor out the constant k and integrate term by term: $k\left[4x - \frac{x^3}{3}\right]_0^2 = 1$ Evaluate at the upper and lower limits: $k\left(\left(4(2) - \frac{2^3}{3}\right) - 0\right) = 1$ $k\left(8 - \frac{8}{3}\right) = 1$.

12 [Intermediate] • Target: Confidence Intervals for Large Unknown-Variance Samples

Given that $\cot\theta = -\frac{3}{4}$ and $\frac{\pi}{2} < \theta < \pi$, find the exact value of $\operatorname{cosec}\theta$.

We use the standard Pythagorean identity relating $\cot\theta$ and $\operatorname{cosec}\theta$: $1 + \cot^2\theta = \operatorname{cosec}^2\theta$. Substitute $\cot\theta = -\frac{3}{4}$: $\operatorname{cosec}^2\theta = 1 + \left(-\frac{3}{4}\right)^2 = 1 + \frac{9}{16} = \frac{25}{16}$. Taking the square root gives: $\operatorname{cosec}\theta = \pm\frac{5}{4}$. Since $\frac{\pi}{2} < \theta < \pi$ (the second quadrant), $\sin\theta > 0$, and therefore $\operatorname{cosec}\theta = \frac{1}{\sin\theta} > 0$. Hence.

13 [Intermediate] • Target: Unbiased Estimators of Population Mean & Variance

Find the exact real solution to the equation $2e^{2x} + 5e^x - 12 = 0$.

We recognize that the equation is quadratic in terms of e^x: $2(e^x)^2 + 5(e^x) - 12 = 0$. Let $u = e^x$. Then the equation becomes: $2u^2 + 5u - 12 = 0$. Factoring the quadratic expression: $(2u-3)(u+4) = 0$. This gives two possible values for u: $u = \frac{3}{2}$ or $u = -4$. Since $u = e^x$: 1. For $e^x = \frac{3}{2}$, taking the natural logarithm of both sides yields: $x = \ln\left(\frac{3}{2}\right)$. 2. For $e^x = -4$, since $e^x > 0$ for all real numbers x, there are no real solu.

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14 **[Intermediate]** • **Target: The Central Limit Theorem (CLT)**

Find the particular solution of the differential equation $\frac{dy}{dx} = xe^{-y}$ given that $y = 0$ when $x = 0$.

To solve the differential equation, separate the variables: $\frac{dy}{e^{-y}} = x\,dx \Rightarrow e^{y}\,dy = x\,dx$. Integrate both sides: $\int e^{y}\,dy = \int x\,dx \Rightarrow e^{y} = \frac{1}{2}x^{2} + c$. Apply the initial condition $y = 0$ when $x = 0$: $e^{0} = \frac{1}{2}(0)^{2} + c \Rightarrow 1 = c$. Substitute $c = 1$ into the solution: $e^{y} = \frac{1}{2}x^{2} + 1$. Taking the natural logarithm of both s.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Confidence Intervals for the Population Mean** • **Cross-Lesson: 15.01 Hypothesis Testing**

Find the exact value of the definite integral $\int_{0}^{\ln 2} e^{2x}\,dx$.

To evaluate the definite integral: 1. Find the antiderivative using the standard standard result $\int e^{kx}\,dx = \frac{1}{k}e^{kx} + C$: $\int e^{2x}\,dx = \frac{1}{2}e^{2x}$ 2. Apply the limits from $x = 0$ to $x = \ln 2$: $\left[\frac{1}{2}e^{2x}\right]_{0}^{\ln 2} = \frac{1}{2}e^{2\ln 2} - \frac{1}{2}e^{2(0)}$ 3. Simplify the exponential terms using the power law of logarithms, $2\ln 2 = \ln(2^{2}) = \ln 4$.

16 **[Challenging]** • **Target: Confidence Intervals for Large Unknown-Variance Samples** • **Cross-Lesson: 2.01 Exponentials & Logarithms**

Relative to a fixed origin O, the points A and B have position vectors $\vec{OA} = \mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$ and $\vec{OB} = 3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$. Find the length of the vector $\vec{AB}$.

To find the vector $\vec{AB}$, subtract the position vector of A from that of B: $\vec{AB} = \vec{OB} - \vec{OA} = (3-1)\mathbf{i} + (2-5)\mathbf{j} + (4-(-2))\mathbf{k} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$. The magnitude (length) of $\vec{AB}$ is: $|\vec{AB}| = \sqrt{2^{2} + (-3)^{2} + 6^{2}} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$. Common misconceptions: - Incorrectly taking $(-3)^{2} = -9$ leads to $\sqrt{4 - 9 + 36} = \sqrt{31}$. - Adding th.

17 **[Challenging]** • **Target: Unbiased Estimators of Population Mean & Variance** • **Cross-Lesson: 3.01 Advanced Trigonometry**

The complex number z satisfies the equation $z(2 - \mathrm{i}) = 7 + 4\mathrm{i}$. Express z in the form $x + \mathrm{i}y$, where x and y are real.

To find z, rearrange the given equation: $z = \frac{7 + 4\mathrm{i}}{2 - \mathrm{i}}$ Multiply the numerator and the denominator by the complex conjugate of the denominator, which is $2 + \mathrm{i}$: $z = \frac{(7 + 4\mathrm{i})(2 + \mathrm{i})}{(2 - \mathrm{i})(2 + \mathrm{i})}$ Expand the denominator: $(2 - \mathrm{i})(2 + \mathrm{i}) = 2^{2} - \mathrm{i}^{2} = 4 - (-1) = 5$ Expand the numerator.

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18 [Challenging] • Target: The Central Limit Theorem (CLT) • Cross-Lesson: 4.01 Differentiation

The first three terms of a geometric progression are 32, 24, and 18. Find the sum to infinity, S_∞, of the progression.

The first term of the geometric progression is $a = 32$. The common ratio r is found by dividing any term by the preceding term: $r = \frac{24}{32} = \frac{3}{4}$ Since $|r| = \left|\frac{3}{4}\right| < 1$, the geometric progression converges and the sum to infinity exists. The sum to infinity is given by: $S_\infty = \frac{a}{1-r}$ Substituting $a = 32$ and $r = \frac{3}{4}$: $S_\infty = \frac{32}{1-\frac{3}{4}} = \frac{32}{\frac{1}{4}} = 32 \times 4 = 128$ Common misconceptions:

19 [Challenging] • Target: Confidence Intervals for the Population Mean • Cross-Lesson: 5.01 Integration

Find $\int x(x^2+1)^5\,dx$ using substitution.

Let $u = x^2 + 1$, $du = 2x\,dx$: $\int x(x^2+1)^5 dx = \frac{x^{12}}{12} + \frac{x^{10}}{2} + \frac{5x^8}{4} + \frac{5x^6}{3} + \frac{5x^4}{4} + \frac{x^2}{2} + C$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

20 [Challenging] • Target: Confidence Intervals for Large Unknown-Variance Samples • Cross-Lesson: 6.01 Sequences & Series

Evaluate $\int_1^3 (x+2)^2\,dx$.

$\int (x+2)^2 dx = \frac{(x+2)^3}{(3)}$. At $x = 3$: base $= 5$, giving 125. At $x = 1$: base $= 3$, giving 27. Result: $\frac{125-27}{3} = \frac{98}{3}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Unbiased Estimators of Population Mean & Variance

What is the unbiased estimator of the population variance σ^2 calculated from a random sample of size n?

a. $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$ ✓

b. $s^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$

c. $s^2 = \frac{1}{n+1}\sum(x_i - \bar{x})^2$

d. $s^2 = \sum(x_i - \bar{x})^2$

Dividing by $n-1$ instead of n removes bias, making $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$ the unbiased estimator of σ^2.

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22 [Foundational] • Target: Unbiased Estimators of Population Mean & Variance

A random sample of 10 observations has $\sum x = 80$ and $\sum x^2 = 724$. Calculate the unbiased estimate of the population variance.

a. 9.33 ✓

b. 8.40

c. 10.50

d. 7.20

$s^2 = \frac{1}{n-1}\left[\sum x^2 - \frac{(\sum x)^2}{n}\right] = \frac{1}{9}\left[724 - \frac{6400}{10}\right] = \frac{1}{9}[724 - 640] = \frac{84}{9} \approx 9.33.$

23 [Foundational] • Target: Unbiased Estimators of Population Mean & Variance

If the biased sample variance is $v = 18$ from a sample of size $n = 25$, find the unbiased estimate s^2.

a. 18.75 ✓

b. 17.28

c. 18.00

d. 19.50

$s^2 = \frac{n}{n-1}v = \frac{25}{24}(18) = 18.75.$

24 [Foundational] • Target: The Central Limit Theorem (CLT)

State the Central Limit Theorem for the distribution of the sample mean $\bar{X}$.

a. **For large n, $\bar{X}$ is approximately normal $N(\mu, \sigma^2/n)$ regardless of the population distribution** ✓

b. For any n, $\bar{X}$ is strictly normal

c. For large n, the population becomes normal

d. $\bar{X}$ has mean $n\mu$ and variance $n\sigma^2$

The CLT states that the sample mean $\bar{X}$ of a random sample from ANY distribution with mean μ and variance σ^2 is approximately normally distributed $N(\mu, \sigma^2/n)$ as $n \to \infty$ (typically $n \geq 30$).

25 [Foundational] • Target: The Central Limit Theorem (CLT)

A population has mean 100 and standard deviation 15. A sample of size $n = 36$ is taken. What is the standard error of the sample mean?

a. 2.5 ✓

b. 15

c. 4.17

d. 0.417

Standard error $\text{SE} = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{36}} = \frac{15}{6} = 2.5.$

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26 [Foundational] • Target: The Central Limit Theorem (CLT)

Using the distribution from the previous question ($\mu = 100$, SE $= 2.5$), find $P(\bar{X} > 105)$.

a. 0.0228 ✓

b. 0.0500

c. 0.0456

d. 0.0100

$z = \frac{105 - 100}{2.5} = 2.0$. $P(\bar{X} > 105) = 1 - \Phi(2.0) = 1 - 0.9772 = 0.0228$.

27 [Foundational] • Target: Confidence Intervals for the Population Mean

A sample of size $n = 64$ from a normal population with known $\sigma = 8$ has sample mean $\bar{x} = 52.0$. Find the 95% confidence interval for μ.

a. [50.04, 53.96] ✓

b. [49.50, 54.50]

c. [50.36, 53.64]

d. [51.00, 53.00]

Margin of error $= z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 1.96 \times \frac{8}{8} = 1.96$. Interval $= 52.0 \pm 1.96 = [50.04, 53.96]$.

28 [Foundational] • Target: Confidence Intervals for the Population Mean

What is the critical value z used to construct a 99% confidence interval for the population mean?

a. 2.576 ✓

b. 1.960

c. 1.645

d. 2.326

For a 99% two-tailed confidence interval, $1 - \alpha = 0.99 \Rightarrow \alpha/2 = 0.005$. $\Phi(2.576) = 0.995$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Confidence Intervals for the Population Mean

How large must a sample be so that the width of a 95% confidence interval for μ is at most 2 units, given $\sigma = 10$?

a. **At least 385** ✓

b. At least 100

c. At least 196

d. At least 250

Width $= 2z\frac{\sigma}{\sqrt{n}} \le 2 \Rightarrow z\frac{\sigma}{\sqrt{n}} \le 1 \Rightarrow 1.96 \times \frac{10}{\sqrt{n}} \le 1 \Rightarrow \sqrt{n} \ge 19.6 \Rightarrow n \ge 19.6^2 = 384.16$. Least integer is 385.

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30 [Intermediate] • Target: Confidence Intervals for Large Unknown-Variance Samples

When σ^2 is unknown but the sample size $n = 100$ is large, what is substituted for σ^2 in calculating the confidence interval?

a. **The unbiased sample variance estimate s^2** ✓

b. The sample range divided by 4

c. The median absolute deviation

d. A default value of 1.0

By the Central Limit Theorem and Slutsky's theorem, when n is large, the unknown population variance σ^2 is replaced by the unbiased sample variance estimate s^2.

31 [Intermediate] • Target: Confidence Intervals for Large Unknown-Variance Samples

A sample of 100 lightbulbs has mean life $\bar{x} = 1200$ hours and unbiased sample variance $s^2 = 2500$. Find the 90% confidence interval for the true mean life.

a. $[1191.8, 1208.2]$ ✓

b. $[1180.0, 1220.0]$

c. $[1190.2, 1209.8]$

d. $[1195.0, 1205.0]$

$s = \sqrt{2500} = 50$. Standard error $= 50/\sqrt{100} = 5$. For 90% confidence, $z = 1.645$. Margin $= 1.645(5) = 8.225$. Interval is $[1191.8, 1208.2]$.

32 [Intermediate] • Target: Confidence Intervals for Large Unknown-Variance Samples

What is the correct probabilistic interpretation of a 95% confidence interval $[L, U]$ for μ?

a. **In repeated sampling, 95% of such constructed intervals will contain the true parameter μ** ✓

b. There is a 95% probability that μ lies in this specific interval

c. 95% of population values fall between L and U

d. The sample mean falls in $[L, U]$ 95% of the time

The parameter μ is a fixed unknown constant; the interval endpoints are random variables. 95% of intervals generated by this procedure over repeated samples will capture μ.

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33 [Intermediate] • Target: Unbiased Estimators of Population Mean & Variance

A sample of size 10 has $\sum x = 420$ and $\sum x^2 = 19424$. Find the unbiased estimate of the population variance.

a. $\frac{1784}{9}$ ✓

b. $\frac{1793}{9}$

c. $\frac{9712}{5}$

d. $\frac{892}{5}$

$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{19424 - 420^2/10}{9} = \frac{1784}{9}$.

34 [Intermediate] • Target: The Central Limit Theorem (CLT)

A sample of size 10 has $\sum x = 350$ and $\sum x^2 = 13030$. Find the unbiased estimate of the population variance.

a. $\frac{260}{3}$ ✓

b. $\frac{263}{3}$

c. 1303

d. 78

$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{13030 - 350^2/10}{9} = \frac{260}{3}$.

35 [Intermediate] • Target: Confidence Intervals for the Population Mean

A sample of size 8 has $\sum x = 392$ and $\sum x^2 = 19402$. Find the unbiased estimate of the population variance.

a. $\frac{97}{4}$

b. $\frac{9701}{4}$

c. $\frac{194}{7}$ ✓

d. $\frac{201}{7}$

$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{19402 - 392^2/8}{7} = \frac{194}{7}$.

36 [Intermediate] • Target: Confidence Intervals for Large Unknown-Variance Samples

A sample of size 8 has $\sum x = 368$ and $\sum x^2 = 17188$. Find the unbiased estimate of the population variance.

a. $\frac{4297}{2}$

b. $\frac{267}{7}$

c. $\frac{260}{7}$ ✓

d. $\frac{65}{2}$

$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{17188 - 368^2/8}{7} = \frac{260}{7}$.

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37 [Intermediate] • Target: Unbiased Estimators of Population Mean & Variance

A sample of size 10 has $\sum x = 120$ and $\sum x^2 = 1672$. Find the unbiased estimate of the population variance.

a. $\frac{116}{5}$

b. $\frac{836}{5}$

c. $\frac{232}{9}$ ✓

d. $\frac{241}{9}$

$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{1672 - 120^2/10}{9} = \frac{232}{9}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The Central Limit Theorem (CLT) • Cross-Lesson: 4.01 Differentiation

A sample of size 7 has $\sum x = 322$ and $\sum x^2 = 15500$. Find the unbiased estimate of the population variance.

a. $\frac{15500}{7}$

b. $\frac{347}{3}$

c. $\frac{344}{3}$ ✓

d. $\frac{688}{7}$

$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{15500 - 322^2/7}{6} = \frac{344}{3}$.

39 [Challenging] • Target: Confidence Intervals for the Population Mean • Cross-Lesson: 5.01 Integration

A sample of size 8 has $\sum x = 376$ and $\sum x^2 = 17978$. Find the unbiased estimate of the population variance.

a. $\frac{153}{4}$

b. $\frac{306}{7}$ ✓

c. $\frac{8989}{4}$

d. $\frac{313}{7}$

$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{17978 - 376^2/8}{7} = \frac{306}{7}$.

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40 [Challenging] • Target: Confidence Intervals for Large Unknown-Variance Samples • Cross-Lesson: 6.01 Sequences & Series

A sample of size 6 has $\sum x = 282$ and $\sum x^2 = 13312$. Find the unbiased estimate of the population variance.

a. $\frac{6656}{3}$

b. $\frac{58}{5}$ ✓

c. $\frac{63}{5}$

d. $\frac{29}{3}$

$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{13312 - 282^2/6}{5} = \frac{58}{5}$.

41 [Challenging] • Target: Unbiased Estimators of Population Mean & Variance • Cross-Lesson: 7.01 Vectors

A sample of size 6 has $\sum x = 234$ and $\sum x^2 = 9262$. Find the unbiased estimate of the population variance.

a. $\frac{68}{3}$

b. $\frac{4631}{3}$

c. $\frac{141}{5}$

d. $\frac{136}{5}$ ✓

$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{9262 - 234^2/6}{5} = \frac{136}{5}$.

42 [Challenging] • Target: The Central Limit Theorem (CLT) • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

A sample of size 5 has $\sum x = 230$ and $\sum x^2 = 10726$. Find the unbiased estimate of the population variance.

a. $\frac{146}{5}$

b. $\frac{73}{2}$ ✓

c. $\frac{75}{2}$

d. $\frac{10726}{5}$

$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{10726 - 230^2/5}{4} = \frac{73}{2}$.

43 [Challenging] • Target: Confidence Intervals for the Population Mean • Cross-Lesson: 9.01 Complex Numbers

A sample of size 6 has $\sum x = 114$ and $\sum x^2 = 2392$. Find the unbiased estimate of the population variance.

a. $\frac{1196}{3}$

b. $\frac{226}{5}$ ✓

c. $\frac{113}{3}$

d. $\frac{231}{5}$

$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{2392 - 114^2/6}{5} = \frac{226}{5}$.

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44 [Challenging] • Target: Confidence Intervals for Large Unknown-Variance Samples • Cross-Lesson: 1.01 Functions: Modulus & Inverse

A sample of size 10 has $\sum x = 220$ and $\sum x^2 = 5098$. Find the unbiased estimate of the population variance.

a. $\frac{86}{3}$ ✓

b. $\frac{2549}{5}$

c. $\frac{129}{5}$

d. $\frac{89}{3}$

$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{5098 - 220^2/10}{9} = \frac{86}{3}$.

45 [Challenging] • Target: Unbiased Estimators of Population Mean & Variance • Cross-Lesson: 10.01 Numerical Methods

A sample of size 8 has $\sum x = 176$ and $\sum x^2 = 4680$. Find the unbiased estimate of the population variance.

a. 101

b. $\frac{815}{7}$

c. 585

d. $\frac{808}{7}$ ✓

$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{4680 - 176^2/8}{7} = \frac{808}{7}$.

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